Heredity: inheritance and variation

Genes, chromosomes and DNA

Traits are characteristics that are passed from parents to their offspring, such as the shape of a seed or the colour of a flower. The instructions for traits are carried by genes. A gene is a segment of DNA, and genes are found on chromosomes inside the nucleus of a cell.

Human body cells contain 46 chromosomes arranged in 23 pairs. One chromosome of each pair comes from the mother and the other comes from the father. Because chromosomes come in pairs, most genes come in pairs too.

Different versions of the same gene are called alleles. For example, a gene for stem length in pea plants has one allele for tall stems and another allele for short stems.

Mendel's pea plant experiments

Gregor Mendel, a monk who worked in the 1800s, studied inheritance by crossing pea plants that showed contrasting traits, such as tall and short stems.

When Mendel crossed pure-breeding tall plants with pure-breeding short plants, all of the offspring in the first generation, called F1, were tall. When he let the F1 plants self-pollinate, about three-fourths of the second generation, called F2, were tall and about one-fourth were short.

From these results Mendel concluded that each plant carries two factors for a trait, which we now call alleles. One allele can hide the effect of the other: the allele that shows is dominant and the hidden one is recessive. The two alleles separate when sex cells (gametes) are formed, so each gamete carries only one allele. This idea is called the law of segregation.

Using a Punnett square

Geneticists write the dominant allele with a capital letter and the recessive allele with the same letter in lowercase. For stem length, T stands for tall and t stands for short.

The combination of alleles an organism has is its genotype. The trait that shows is its phenotype. An organism with two identical alleles, TT or tt, is homozygous. An organism with two different alleles, Tt, is heterozygous. A heterozygous pea plant is tall, because T is dominant.

A Punnett square shows all the possible allele combinations from a cross. In a cross between two heterozygous plants (Tt x Tt), the offspring genotypes are TT, Tt, Tt and tt. The genotypic ratio is 1 TT : 2 Tt : 1 tt, and the phenotypic ratio is 3 tall : 1 short. The probability that any one offspring is short is 1 in 4, or 25 percent.

Patterns beyond simple dominance

Not every trait follows simple dominance. In incomplete dominance, the heterozygous offspring shows a phenotype in between the two parents. When a red-flowered four o'clock plant is crossed with a white-flowered one, the offspring have pink flowers.

In codominance, both alleles are fully expressed. Roan cattle have both red hairs and white hairs. In human ABO blood types, the A and B alleles are codominant, so a person who inherits both has type AB blood.

Some genes have more than two alleles in a population. The ABO blood type gene has three alleles: IA, IB and i. The i allele is recessive, so a person with type O blood has the genotype ii.

Sex-linked traits are controlled by genes on the X chromosome. Red-green colour blindness is more common in males because males have only one X chromosome, so a single recessive allele on that X chromosome is enough for the trait to appear.